

DexRoam: Learning Mobile Bimanual Dexterous Manipulation from Egocentric Whole-Body Human Demonstrations

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Project Page: dexroam.github.io

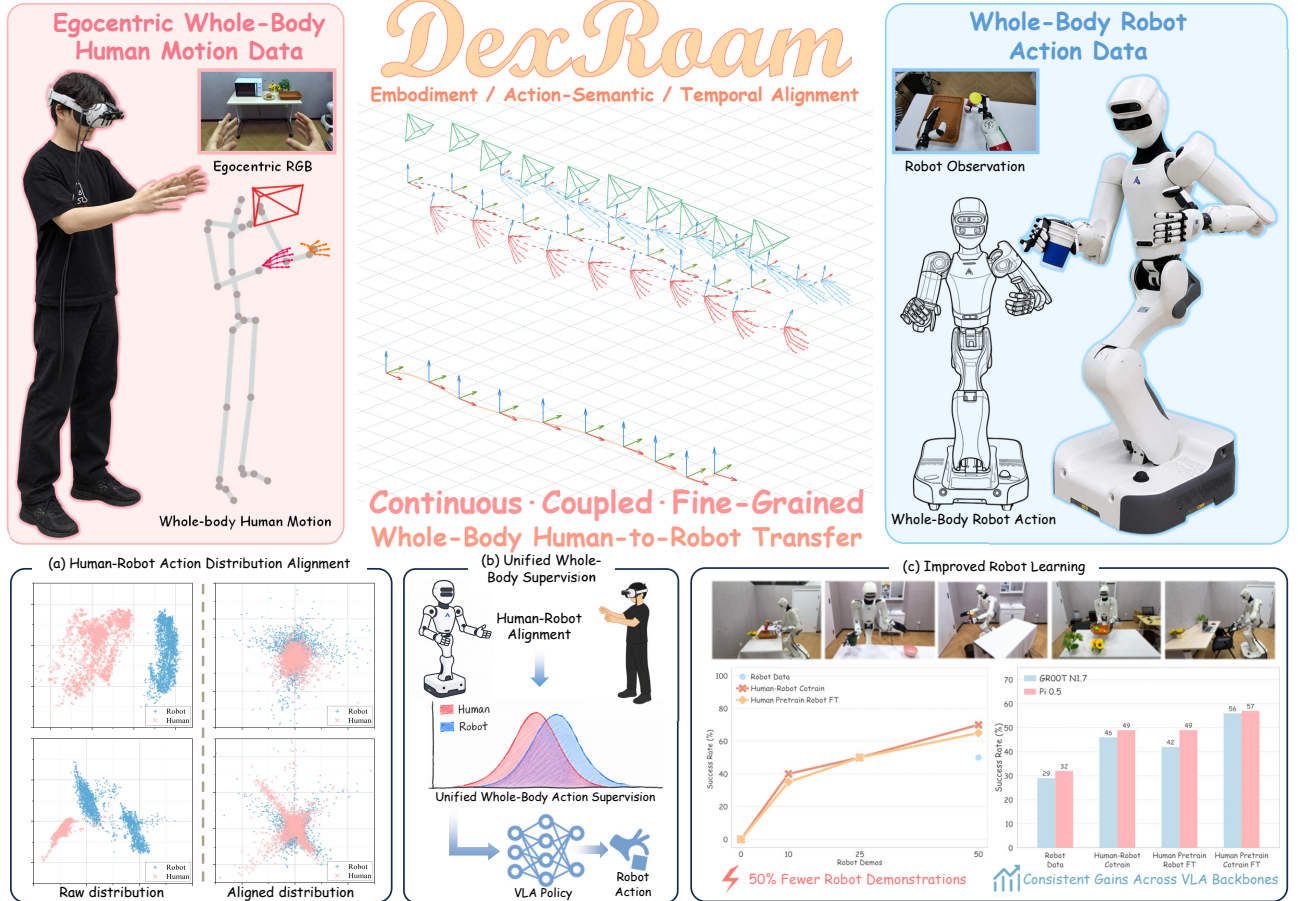


Figure 1. Overview of the DexRoam framework. DexRoam unlocks motion-level human data for mobile bimanual dexterous manipulation by transforming egocentric whole-body demonstrations into continuous, coupled robot-compatible action supervision. Through explicit embodiment, action-semantic, and temporal alignment, DexRoam substantially closes the human-robot action distribution gap and consistently improves policy learning across VLA backbones, while reducing robot demonstration requirements.

Abstract—Mobile bimanual dexterous manipulation requires continuous coordination of locomotion, whole-body motion, and finger-level dexterity within a single trajectory, creating a severe robot demonstration bottleneck. Egocentric human demonstrations offer a scalable alternative, but prior approaches ease the transfer by simplifying human motion, discarding exactly the fine-grained, coupled structure such tasks depend on. We present DexRoam, a complete system for learning mobile bimanual dexterous manipulation from human demonstrations, in which whole-body motion remains continuous and coupled throughout the human-to-robot transfer process. To enable scalable collection of whole-body human manipulation demonstrations, we develop a tracker-free capture system using only a consumer VR headset and a head-mounted stereo camera, without external cameras

or motion trackers. We then perform three explicit alignment stages—embodiment, action-semantic, and temporal—to map captured motion into the robot action space, preserving fine-grained whole-body motion and allowing human and robot demonstrations to be jointly learned by standard VLA policies. Real-world experiments with different VLA backbones show that human demonstrations consistently improve policy learning across training paradigms, raising average success from 29% to 56% on GR00T N1.7 and from 32% to 57% on $\pi_{0.5}$, while matching robot-only training with half the robot demonstrations. Ablations confirm that each alignment stage is necessary. These results highlight the potential of human demonstrations for scalable whole-body mobile manipulation with preserved fine-grained motion structure.

I. INTRODUCTION

Mobile bimanual dexterous manipulation requires robots to simultaneously perform locomotion, torso adjustment, bimanual coordination, active vision, and finger-level dexterous interaction within a single trajectory. Unlike isolated manipulation or locomotion, these tasks require continuous coordination among multiple body components throughout execution. Training such systems therefore demands large-scale, high-quality whole-body action data. However, collecting robot demonstrations at this scale remains challenging as robot embodiments become increasingly complex. Existing mobile manipulation platforms, such as Human Support Robot [1] and Stretch [2], provide important foundations for embodied intelligence, while recent whole-body teleoperation systems extend data collection toward bimanual and humanoid manipulation through bilateral interfaces, kinematic-matched platforms, and wearable devices [3]–[6]. Nevertheless, these approaches require increasingly specialized hardware interfaces, making large-scale data collection for mobile bimanual dexterous robots expensive and difficult to scale.

Egocentric human demonstrations provide a scalable alternative for alleviating the robot data bottleneck. Large-scale egocentric datasets have revealed rich human activity priors [7]–[9], while recent systems enable collecting robot-relevant human demonstrations through scalable interfaces [10]–[13]. Building on these advances, recent works demonstrate that human demonstrations can directly improve robot policy learning for dexterous manipulation [14]–[16], bimanual manipulation [17], and vision-language-action models trained from egocentric data [18]. However, directly transferring the full human motion space to robots remains challenging due to differences in morphology, control interfaces, and execution timescales between humans and robots.

To reduce this transfer difficulty, prior human-to-robot works often simplify or restructure human motion representations before policy learning. Instead of directly transferring the complete human motion space, these approaches introduce structured intermediate representations that reduce embodiment discrepancies and make robot learning more tractable. Recent mobile whole-body manipulation systems adopt different forms of such representation transformations, including simplified locomotion and hand representations [19], phase-based decomposition of navigation and manipulation [20], and controller-assisted whole-body execution through task-level targets [21]–[24]. Other approaches avoid directly transferring human actions and instead leverage human videos for representation learning or foundation model pretraining [25]–[28].

These representation transformations reduce the complexity of human–robot transfer, but they inherently introduce a trade-off between transferability and preserving human motion information. This trade-off is often acceptable for robots with limited action complexity and weak whole-body coupling, but becomes restrictive for mobile bimanual dexterous manipulation, where successful execution requires continuous coordination among locomotion, upper-body motion, and dexterous interaction. Specifically, existing approaches face several limitations. First, compressing continuous human

motion into discrete or low-dimensional representations can remove fine-grained motion variations, preventing policies from fully exploiting the structure in human demonstrations. Second, decomposing long-horizon behaviors into independent phases can weaken dependencies between different stages of execution, making it difficult to jointly learn locomotion and upper-body manipulation within a unified policy. For example, separating navigation and manipulation limits the ability to perform tasks requiring tightly coupled arm–base coordination. Finally, in mobile bimanual manipulation, the motion of one body component often changes the feasible workspace and control constraints of others. Representing the base, torso, arms, and hands as independent control objectives ignores these interactions, limiting the ability to learn the whole-body coordination required for mobile dexterous manipulation.

This raises the central question of this work: **Can human demonstrations be transferred to robots while resolving embodiment mismatch and preserving the fine-grained motion structure required for complex whole-body behaviors?**

We present **DexRoam** (Fig. 1), a complete system that answers this question by learning mobile bimanual dexterous manipulation from human demonstrations end to end, spanning portable egocentric capture, explicit whole-body human-to-robot alignment, and real-robot policy learning. The system is built around a single design principle: the continuous, coupled structure of whole-body motion is carried from the demonstrator’s body to the robot’s action chunks without discretized locomotion primitives, phase decomposition, or independent per-component control. Each stage is designed so that the continuous, coupled structure of whole-body motion survives from the demonstrator’s body to the robot’s action.

The first stage is a **tracker-free egocentric whole-body capture system**. It recovers whole-body human motion together with egocentric video using only a consumer VR headset and a head-mounted stereo camera, requiring no external cameras, environment markers or body-worn trackers. The camera stream is displayed inside the headset in real time, so the demonstrator’s view during collection is exactly the observation the policy later sees. This keeps collection portable and scalable, while providing raw whole-body motion in human space as the input to the rest of the system.

The second stage is **explicit human-to-robot alignment**, which brings this motion into the robot’s action space without compressing it. Three mismatches are resolved in turn. *Embodiment mismatch*, from structural differences between humans and robots, is handled by root-centric whole-body retargeting. *Action-semantic mismatch*, between human motion representations and robot control interfaces, is handled by robot-centric relative action construction. *Temporal mismatch*, from the different execution timescales of humans and robots, is handled by task-progress temporal resampling. After alignment, human and robot demonstrations occupy the same observation–action space, so a standard VLA policy trains on both jointly, without human-specific action heads, and directly predicts complete whole-body action chunks.

We validate the full pipeline on a real mobile bimanual dexterous robot across five manipulation tasks, two VLA

backbones, and varying robot-data budgets. Transferred human trajectories replay on the robot with centimeter-level error and no IK failures, and alignment substantially reduces the human–robot action distribution gap. Training on the aligned data improves policy performance consistently across tasks and backbones, and matches robot-only training with half the robot demonstrations. Ablations show that removing action-semantic or temporal alignment substantially degrades the policy, confirming that each stage is necessary for the pipeline to work. Our contributions are:

- *DexRoam*, a complete working system for learning mobile bimanual dexterous manipulation from human demonstrations, integrating portable egocentric capture, explicit whole-body alignment, and real-robot policy learning under a single continuous, coupled action interface.
- A *tracker-free egocentric whole-body capture system* that records full-body motion, hand articulation, and egocentric stereo video with only a consumer VR headset and a head-mounted camera, with in-headset streaming that keeps the demonstrator’s view consistent with the policy observation.
- An *explicit three-stage alignment* resolving embodiment, action-semantic, and temporal mismatches while preserving continuous whole-body motion structure, so that human and robot data are interchangeable for a standard VLA policy.
- *Extensive real-world evaluation* across five tasks, two VLA backbones, and different robot-data budgets, covering trajectory executability, distribution alignment, policy improvement, data efficiency, and ablations.

II. PROBLEM DEFINITION

We study the transfer of human demonstrations to mobile bimanual dexterous robots. We define the robot whole-body action space as $\mathcal{A}$, spanning the mobile base, torso, dual arms, active head, and dexterous hands, and require that the transferred demonstrations satisfy a *fine-grained whole-body action interface* with three properties: **(1) Continuous action space.** All action dimensions remain continuous rather than being discretized. **(2) Temporally continuous prediction.** The complete action sequence is continuously predicted by a single policy, without phase switching at deployment. **(3) Coupled whole-body prediction.** Actions of all body components are jointly modeled in a unified policy output space, rather than being treated as independent control objectives during deployment. Under this setting, we study how to effectively transfer human demonstrations to mobile bimanual dexterous robots while satisfying the above action interface constraints.

The training corpus consists of two sources: robot demonstrations $\mathcal{D}_{\text{robot}}$ collected on the target embodiment, whose actions already lie in $\mathcal{A}$, and human demonstrations $\mathcal{D}_{\text{human}}$, whose motions reside in human space. We therefore require a human-to-robot transformation $\mathcal{R}$ satisfying two conditions:

$$\mathcal{R}(\mathcal{D}_{\text{human}}) \subset \mathcal{S}, \quad p_{\mathcal{R}(\mathcal{D}_{\text{human}})} \approx p_{\mathcal{D}_{\text{robot}}},$$

where $\mathcal{S}$ is the observation–action space of the target embodiment, whose action component $\mathcal{A}$ satisfies the three properties above. The first condition makes both sources

consumable by a single policy without human-specific action heads; the second ensures the transferred data do not pull the policy away from the action distribution it encounters at deployment. Only under both conditions can $\mathcal{R}(\mathcal{D}_{\text{human}})$ and $\mathcal{D}_{\text{robot}}$ be trained on jointly under a single objective. We define the policy as $\pi_{\theta}(\mathbf{A}_t \mid \mathbf{I}_t, \mathbf{P}_t, \ell)$, where the policy receives an egocentric visual observation $\mathbf{I}_t$, a proprioceptive state history $\mathbf{P}_t$, and a language instruction ℓ , and outputs a continuous whole-body action chunk $\mathbf{A}_t \in \mathcal{A}$. After alignment, transformed human demonstrations join the training corpus $\mathcal{D}$ alongside robot demonstrations, enabling joint learning under the same policy and action interface.

III. DEXROAM

In this section, we present DexRoam. Sec. III-A introduces the data collection system, Sec. III-B presents the three-stage alignment addressing embodiment, action-semantic, and temporal mismatches, and Sec. III-B4 describes policy learning under the unified observation–action interface.

A. Hardware Platform and Data Collection System

1) **Hardware platform:** The Atribot–XHand platform comprises an omnidirectional mobile base, an actuated torso, two 7-DoF arms, a 2-DoF active head, and two 12-DoF XHand dexterous hands (Fig. 2a, right), supporting coordinated whole-body mobile dexterous manipulation.

2) **Robot demonstration collection:** We build a whole-body teleoperation system (Fig. 2a) for collecting robot demonstrations on the target mobile bimanual dexterous platform. The Meta Quest 3S captures upper-body poses, VR controllers provide wrist pose commands, and MANUS Pro gloves capture hand motions, with these inputs jointly mapped to the robot base, end-effectors, and XHand joints, enabling coupled whole-body mobile dexterous teleoperation. The system adopts a single-operator control paradigm, where the operator simultaneously controls robot locomotion and whole-body manipulation, eliminating the need for an additional operator dedicated to base navigation. Since wearing gloves makes conventional keyboard or button-based interaction inconvenient, we further design a gesture-based interface for episode state management, allowing the operator to start, stop, save, and discard episode directly through hand gestures.

3) **Tracker-free egocentric whole-body data collection:** Human demonstrations are captured with only a Meta Quest 3 headset and a head-mounted ZED Mini stereo camera (Fig. 2b), requiring no external motion-capture cameras, no environment markers and no body-worn trackers. The Meta Quest 3 captures whole-body motion through WebXR [29], including 6-DoF headset pose, 33 body-joint poses, and 25 hand landmarks per hand, which together form the whole-body human state $\mathbf{h}_t$. Meanwhile, the ZED Mini records egocentric stereo RGB images with timestamps, providing first-person visual observations for each demonstration. Fig. 2b (right) shows a captured whole-body mobile manipulation trajectory, in which head pose, hand keypoints, and the full-body skeleton evolve continuously as the demonstrator moves through the



Fig. 2. **DexRoam data collection system.** (a) Follow-style teleoperation lets one operator drive base, torso, arms, head, and dexterous hands simultaneously via a Meta Quest 3S, VR controllers, and MANUS Pro gloves, with gesture-based episode management. (b) Tracker-free egocentric capture uses a Meta Quest 3 and a head-mounted ZED Mini to record 6-DoF head pose, 33 body joints, 25 joints per hand, and timestamped egocentric stereo RGB, with no external cameras, marker or trackers. Real-time in-headset video streaming keeps the demonstrator’s view identical to the policy observation.

scene. A gesture-based interface is also designed for episode management, allowing the operator to start, stop, save, and discard recordings directly through gestures.

The key design choice of the capture system is *egocentric video streaming*: the ZED Mini stream is displayed inside the headset in real time (Fig. 2b), so what the operator sees during collection is exactly the egocentric observation used later for policy training. This keeps visual observation, visual attention, and motor intent consistent throughout an episode. Such consistency is important for mobile dexterous manipulation, where the viewpoint continuously changes as the robot moves its base, body, and head.

B. Human-to-Robot Alignment

DexRoam instantiates the human-to-robot transformation $\mathcal{R}$ through three explicit alignment steps (Fig. 3), which resolve embodiment, action-semantic, and temporal mismatches between human demonstrations and robot execution spaces.

1) **Embodiment Alignment:** Humans and robots have different kinematic trees, so neither joint angles nor global poses can be copied directly into an executable action. To reduce dependence on global coordinate frames and preserve body-relative motion structure, we introduce a time-varying pelvis-centric root frame $\mathcal{F}_H$ to align human motion with the robot embodiment. The origin of $\mathcal{F}_H$ is the orthogonal projection of the pelvis onto the ground plane (*Projection* in Fig. 3a); its forward axis is the horizontal projection of the pelvis’s local forward direction, and its vertical axis points opposite to gravity. Let $\mathbf{z}_H = -\widehat{\mathbf{w}}\mathbf{g}$ denote the upward axis of the human root frame, where $\mathbf{w}\mathbf{g}$ is the gravity vector expressed in the world frame. The ground-plane projection operator is defined as $\Pi_G = \mathbf{I} - \mathbf{z}_H \mathbf{z}_H^\top$. Assuming that the world origin lies on the ground plane, the pose of $\mathcal{F}_H$ is defined as:

$$\mathbf{x}_t = \Pi_G \widehat{\mathbf{w}}\mathbf{f}_{\text{pelvis},t}, \quad \mathbf{y}_t = \mathbf{z}_H \times \mathbf{x}_t, \\ \mathbf{w}\mathbf{T}_{H,t} = \begin{bmatrix} \mathbf{x}_t & \mathbf{y}_t & \mathbf{z}_H & \Pi_G \mathbf{w}\mathbf{p}_{\text{pelvis},t} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Aligned with the robot base motion plane, $\mathcal{F}_H$ establishes a human-centric reference frame shared by locomotion and manipulation. It serves two purposes: transferring the human locomotion trajectory to the robot chassis motion and representing upper-body movements as body-relative trajectories, reducing dependence on global scene coordinates. Specifically, torso motion, head orientation, and wrist trajectories are expressed in $\mathcal{F}_H$ through

$${}^H\mathbf{T}_{J,t} = ({}^W\mathbf{T}_{H,t})^{-1} {}^W\mathbf{T}_{J,t},$$

where $J \in \{\text{chest}, \text{head}, \text{l-wrist}, \text{r-wrist}\}$. The resulting relative trajectories are then converted into robot commands: the human root trajectory is mapped to the chassis motion, the chest pose is used as the torso target, and the left/right wrist poses are transformed into arm end-effector targets. For the head, although its pose is represented in $\mathcal{F}_H$, only the yaw and pitch components are extracted as the 2-DoF head command due to the robot’s mechanical constraints. Fig. 3b illustrates the resulting per-component trajectories in robot space. For the dexterous hands, we adopt dex-retargeting [30] to retarget human hand poses into the 12-DoF XHand commands, bridging the morphology gap between human and robot hands.

2) **Action-Semantic Alignment:** Retargeting yields poses, not actions: the goal is not merely to reach the robot’s pose space, but to construct targets carrying the same semantics as robot control. Absolute poses fail this test, as they still encode demonstrator-specific configuration such as height, shoulder width, arm length, and initial posture. We therefore express continuous motion as relative transformations in each body component’s local frame, which specify how a component moves from where it currently is rather than where it should

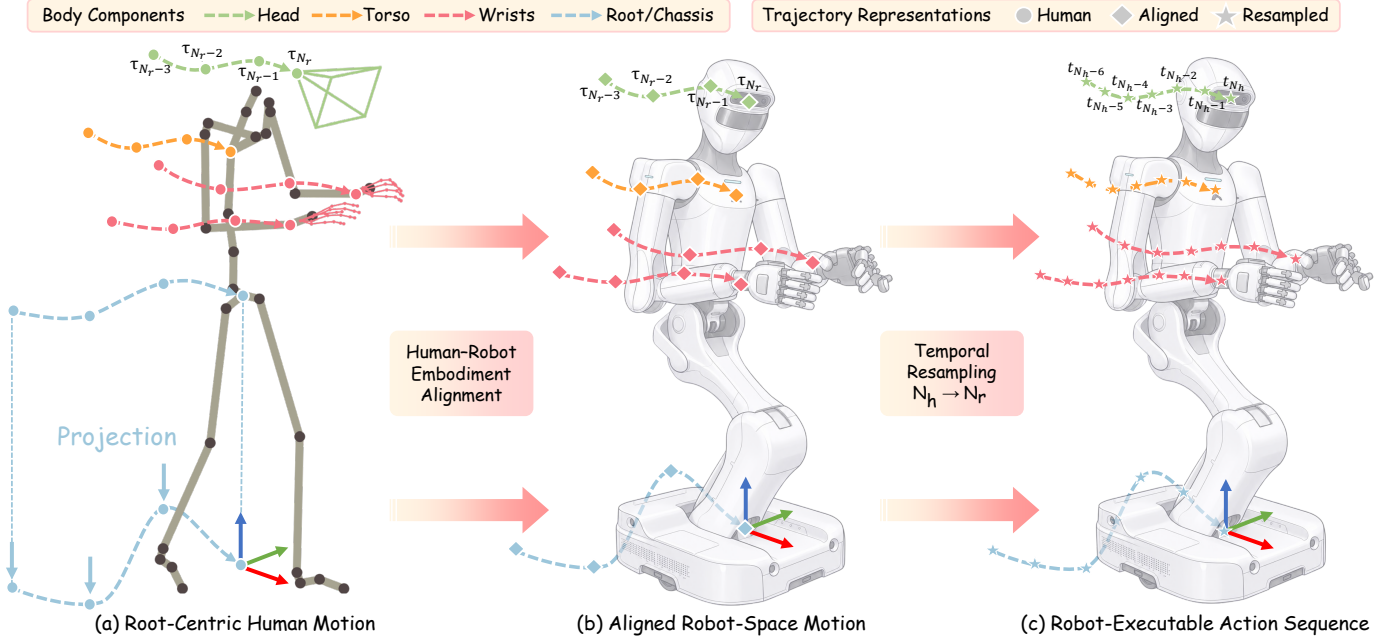


Fig. 3. **Explicit human-to-robot alignment in DexRoom.** (a) Human motion is expressed in a pelvis-centric root frame projected onto the ground plane, yielding root/chassis, torso, wrist, and head trajectories free of global scene coordinates. (b) Embodiment alignment maps each component to its robot counterpart, and action-semantic alignment re-expresses the results as robot-centric relative transformations. (c) Temporal alignment resamples N_h human steps into N_r waypoints evenly spaced in task progress, matching the execution timescale of robot demonstrations.

be in absolute coordinates, and are thus independent of the configuration in which the demonstration was recorded. For upper-body components, poses are already represented in the human root frame $\mathcal{F}_H$, we compute the relative motion for $J \in \{\text{chest, l-wrist, r-wrist}\}$ as

$$\Delta \mathbf{T}_t^J = ({}^H\mathbf{T}_{J,t})^{-1} {}^H\mathbf{T}_{J,t+1}$$

The resulting local transformations describe how each component evolves from its current state and preserve transferable motion patterns such as reaching, lifting, and retracting.

For locomotion, the root frame encodes the human movement. We compute the relative transformation between consecutive root frames and represent it as planar motion:

$$\Delta \mathbf{T}_t^H = ({}^W\mathbf{T}_{H,t})^{-1} {}^W\mathbf{T}_{H,t+1}, \Delta \xi_t^H = [\Delta x_t, \Delta y_t, \Delta \psi_t]^\top$$

Since $\mathcal{F}_H$ is constructed on the ground plane with a gravity-aligned vertical axis, its relative motion is inherently planar. Compared with absolute trajectories, this representation removes dependence on scene coordinates and initial positions while providing explicit locomotion semantics, enabling continuous coupling between base movement and whole-body manipulation. The 2-DoF head command and the 12-DoF hand command already lie in robot joint space and are used directly.

3) Temporal Alignment: We align demonstrations by task progress rather than wall-clock time. Humans complete the same task faster and in fewer steps than the robot, so the two advance task progress at different rates per policy step. We resample each retargeted human trajectory into $\bar{N}_r$ evenly spaced waypoints in normalized task progress $s \in [0, 1]$, where $\bar{N}_r$ is the mean robot episode length for that task:

$$\tilde{T}_k = \text{Interp}(T_{1:N_h}, s_k), s_k = \frac{k-1}{\bar{N}_r-1}, 1 \leq k \leq \bar{N}_r$$

with N_h the original human trajectory length. After resampling, the two sources advance the same fraction of task progress per policy step, as illustrated in Fig. 3c. The resampled waypoints preserve the original spatial trajectory while matching the temporal distribution of robot executions. Positions are interpolated linearly and rotations by SLERP, while images and discrete signals use nearest-neighbor matching; base yaw is unwrapped before interpolation to avoid discontinuities around $\pm\pi$. Resampling only adjusts the temporal discretization of the trajectories and does not modify the robot control frequency or action chunk size.

After alignment, human and robot demonstrations become interchangeable under the same policy interface. The resulting action representation satisfies the three properties defined in Sec. I. First, all actions remain continuous without discretization. Second, the complete whole-body trajectory is predicted by a single policy without phase-based execution. Third, all body components are jointly represented in one unified action space. This unified interface enables human demonstrations to retain their fine-grained motion structure while becoming executable in the robot action space.

4) Policy Learning: Because aligned human data and robot data share the same observation-action space, they can be directly consumed by a standard VLA policy without human-specific action heads or input branches. We instantiate $\pi_\theta(A_t|I_t, P_t, \ell)$ with two VLA backbones, GR00T N1.7 [31] and $\pi_{0.5}$ [32]. Each policy takes egocentric RGB observations, proprioceptive state history, and language instructions as inputs, and predicts whole-body action chunks. We design three strategies for incorporating aligned human demonstrations into policy learning: **Human-Robot Cotrain**, jointly training on aligned human and robot demonstrations; **Human Pretrain + Robot FT**, pretraining on aligned human demon-



Fig. 4. **Real-world mobile bimanual dexterous manipulation tasks.** Rollouts of the five evaluation tasks, one per row: Pick Chips Can, Throw Trash, Deliver Fruit, and Push Chair & Close Laptop. They span fine-grained dexterity, base-manipulation coupling, and long-horizon whole-body interaction.

strations and finetuning on robot demonstrations only; and **Human Pretrain + Cotrain FT**, pretraining on aligned human demonstrations and finetuning on the human-robot mixture. All strategies use the same observation-action interface and downstream finetuning budget.

IV. EXPERIMENTS

A. Experimental Setup

1) **Tasks:** We evaluate DexRoam on five tasks covering complementary mobile manipulation capabilities (Fig. 4). **Pick Chips Can** requires the robot to navigate to a table, grasp a chips can, and place it in a tray, emphasizing precise reaching and finger-level grasping. **Pour Water** requires grasping a cup and pouring into a target container, testing continuous wrist-hand coordination. **Throw Trash** requires grasping an object, moving to a bin, and releasing it accurately; the object must remain stable during locomotion, while successful disposal depends on precise coordination between base positioning and manipulation. **Deliver Fruit** requires carrying a basket bimanually to a target workstation, stressing bimanual transport and coordinated turning during locomotion. **Push Chair & Close Laptop** chains long-range navigation, contact-rich chair pushing, and laptop closing, testing long-horizon execution together with continuous base-upper-body coordination and fine manipulation.

2) **Evaluation protocol:** We evaluate DexRoam at three complementary levels: trajectory executability, human-robot action distribution alignment, and policy performance.

Trajectory replay. We directly replay transferred human trajectories on the real robot and report trajectory completion, IK failure rate, and positional and rotational replay errors for the torso, arms, and head.

Action distribution alignment. We further examine whether the alignment process brings human-derived actions closer to real robot actions. For each task, we compare the human and robot whole-body action distributions at the retargeted-absolute stage and after full alignment. We use Maximum Mean Discrepancy (MMD) and Sliced Wasserstein Distance (SWD) for quantitative comparison and PCA for qualitative visualization. Detailed normalization, metric computation, and visualization protocols are provided in the Appendix.

Policy evaluation. We evaluate all four training settings described in Sec. IV-A on all five tasks, with 20 real-world trials per task. Success rate (SR) captures whether the final task goal is achieved, but provides limited information about partial progress, particularly for mobile manipulation tasks. We therefore additionally use task completion score (TCS), which decomposes each task into ordered stages and measures the fraction of stages completed. Detailed TCS criteria are provided in the Appendix. We report both SR and TCS

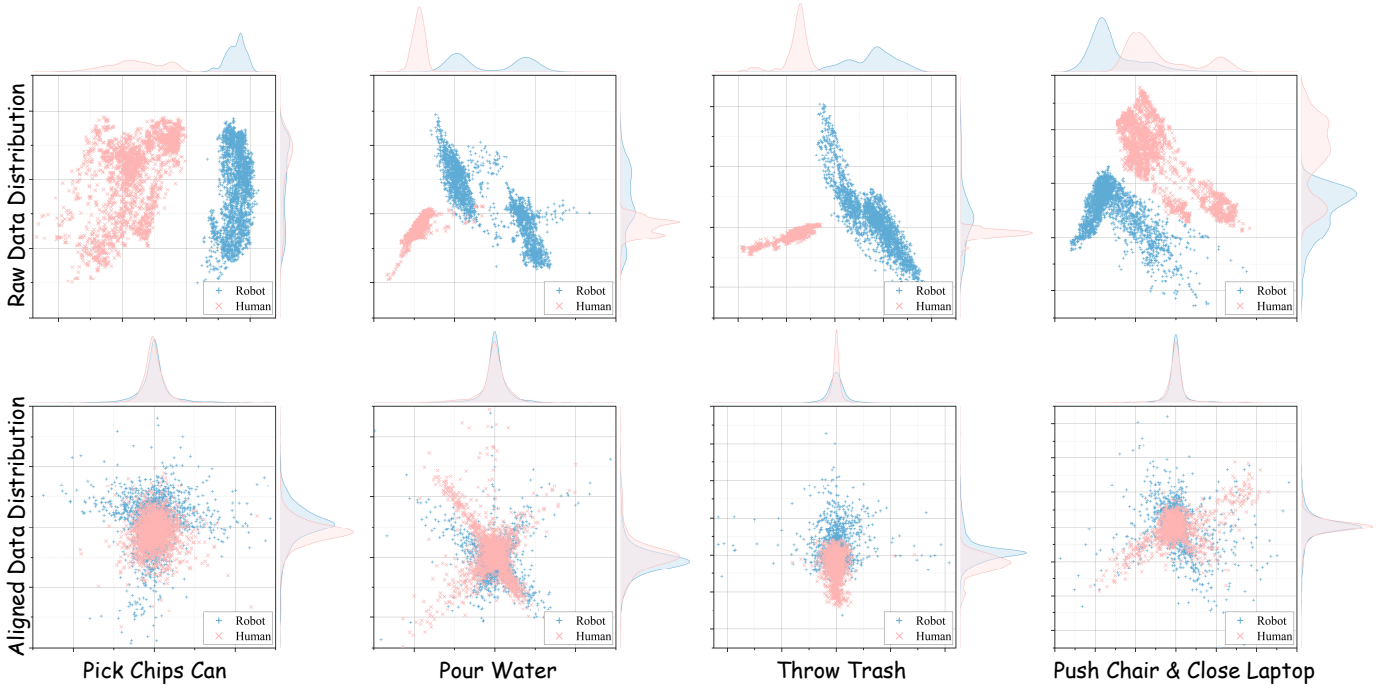


Fig. 5. **Human-robot action distribution before and after alignment.** PCA visualization of whole-body actions on representative tasks. Top: retargeted absolute human actions and robot actions before action-semantic and temporal alignment, which occupy clearly separated regions. Bottom: fully aligned robot-centric relative actions, where the two distributions largely overlap, with marginals shown along each axis.

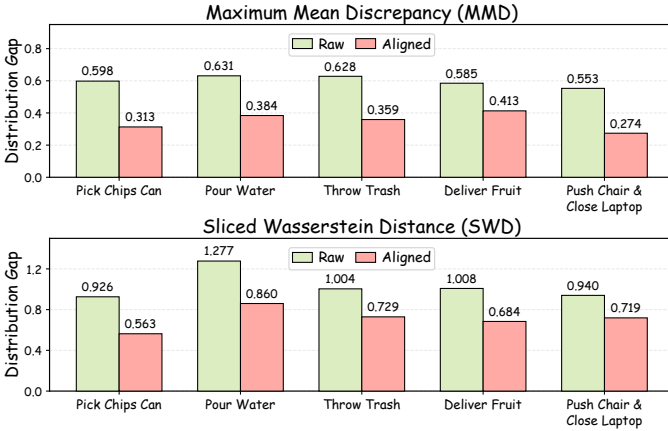


Fig. 6. **Quantitative human-robot action distribution alignment.** MMD and SWD between human-derived and robot whole-body actions on all five tasks, lower values indicate closer distributions. Alignment reduces both metrics on every task, by 41.7% in MMD and 31.1% in SWD on average.

for downstream policy evaluation. Based on these evaluation metrics, we investigate when, how, and why aligned human demonstrations benefit mobile bimanual dexterous manipulation. Specifically, we study four questions:

- **Q1: Transfer Quality.** How effectively does human-to-robot transfer convert human motion into executable and distribution-aligned robot training data?
- **Q2: Policy Learning.** Does aligned human supervision improve policy learning, and which training paradigm leverages it most effectively?
- **Q3: Data Efficiency.** How does aligned human data improve robot learning with limited robot demonstrations?
- **Q4: Alignment Analysis.** How do different alignment choices affect fine-grained human-to-robot transfer?

TABLE I: **Real-robot replay of transferred human trajectories.** Position and rotation errors of retargeted human motions across different body components, measured in millimeters (mm) and degrees ($^{\circ}$).

Component	Metric	Pick Chips Can	Pour Water	Throw Trash	Average
Torso	Position	13.49	11.82	12.31	12.54
	Rotation	2.32	1.64	1.57	1.84
Left Arm	Position	3.75	4.27	4.02	4.01
	Rotation	0.51	0.58	0.57	0.55
Right Arm	Position	5.30	5.93	7.67	6.30
	Rotation	0.70	0.84	1.22	0.92
Head	Rotation	1.22	1.36	1.53	1.37
Completed		5/5	5/5	5/5	15/15

TABLE II: **Tracker-free vs. tracker-based capture.** Retargeting accuracy averaged over 15 trajectories across three tasks. Position and rotation errors are reported in millimeters and degrees, respectively.

Human Motion Capture	Torso		Left Arm		Right Arm		Head
	Pos.	Rot.	Pos.	Rot.	Pos.	Rot.	
Tracker-free (Ours)	12.54	1.84	4.01	0.55	6.30	0.92	1.37
PICO + Motion Trackers	13.16	2.06	4.12	0.57	5.63	0.81	1.23

B. Q1: Transfer Quality

1) **Action distribution alignment.** We examine whether the transferred human supervision is compatible with the action distribution of the real robot. Across all five tasks, alignment consistently reduces both distribution metrics, decreasing MMD by **41.7%** and SWD by **31.1%** on average (Fig. 6). The PCA visualization in Fig. 5 shows the same qualitative trend, with substantially greater overlap between human-derived and robot actions after alignment. Together with the replay results, these findings show that DexRoam produces human supervision that is not only executable on the target embodiment, but also better matched to real robot

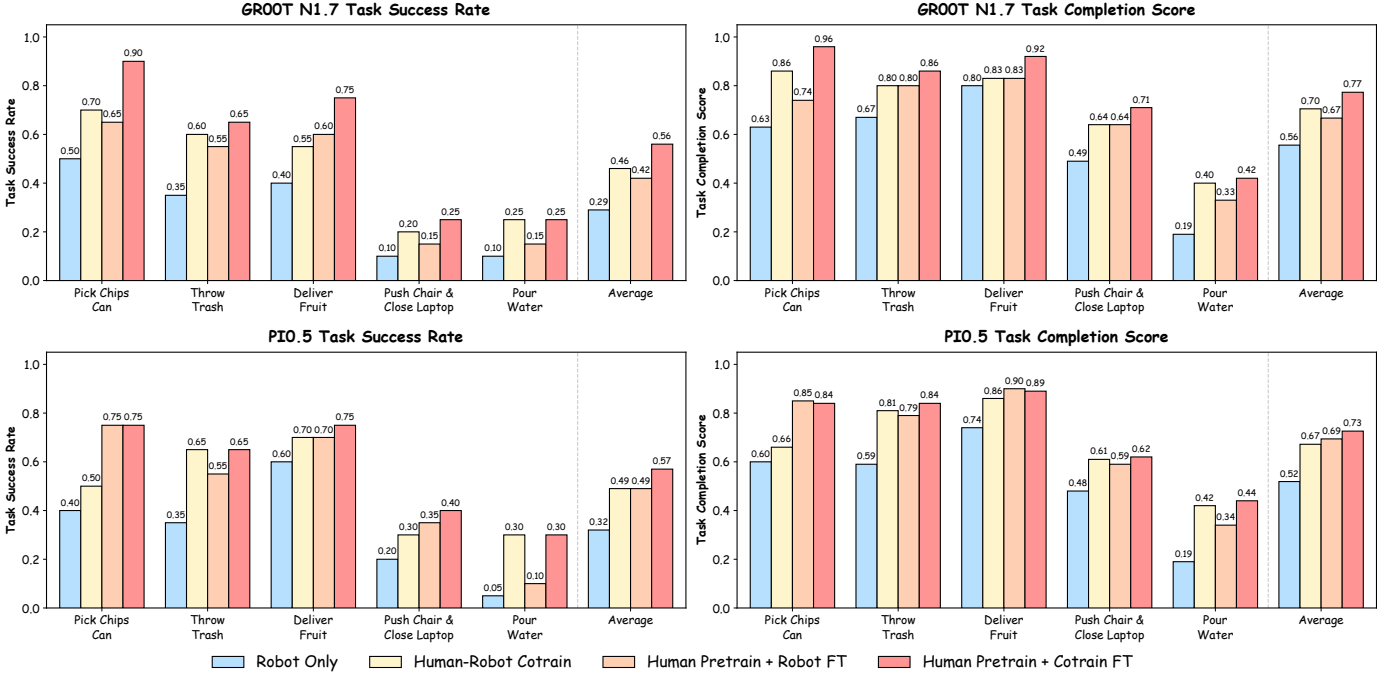


Fig. 7. **Main policy learning results.** Main policy learning results. Success rate (SR, left) and task completion score (TCS, right) of Robot-only and three human-assisted training strategies on five real-world tasks, for GR00T N1.7 (top) and $\pi_{0.5}$ (bottom). All three human-assisted strategies improve over Robot-only on every task and both backbones, with Human Pretrain + Cotrain FT best or tied-best on every task-backbone combination.

actions.

2) **Real-robot executability.**: All 15 transferred human trajectories replay successfully on the real robot with no IK failures (Tab. I). Across the three tasks, positional replay errors remain at approximately the centimeter scale for the torso and millimeter scale for both arms, while rotational errors remain within a few degrees across all evaluated body components. Despite the different whole-body motion patterns involved, the transferred trajectories can therefore be consistently executed by the target robot, supporting their use as robot supervision.

3) **Tracker-free capture fidelity.**: Since tracker-free human motion capture is a key component of our data collection system, we further examine whether this lightweight design compromises the quality of the transferred trajectories. For comparison, we construct a tracker-based counterpart using a PICO headset with five Motion Trackers and evaluate it under the same replay protocol. Both systems successfully replay all trajectories and achieve comparable replay errors across body components, with no systematic performance gap (Tab. II). This suggests that our tracker-free system substantially reduces sensing cost and wearable burden while preserving the quality required for subsequent human-to-robot transfer.

C. Q2: Policy Learning

Relative to Robot-only, all three human-assisted training paradigms improve both SR and TCS across both VLA backbones and all five tasks (Fig. 7). In terms of average SR, the three paradigms provide gains of **+16**, **+11**, and **+27 percentage points** on GR00T N1.7, and **+17**, **+17**, and **+25 percentage points** on $\pi_{0.5}$. The same positive trend holds across individual tasks, indicating that the benefit of aligned human supervision is not tied to a particular training paradigm,

policy backbone, or manipulation setting. This remains true on Push Chair & Close Laptop, the longest-horizon and most tightly coupled task, where the best human-assisted setting improves success by **15 percentage points** on GR00T N1.7 and **20 percentage points** on $\pi_{0.5}$. Together, these results show that aligned human supervision provides a broadly useful learning signal for mobile bimanual dexterous manipulation.

The magnitude of the gain, however, depends on how the human supervision is used. Human Pretrain + Cotrain FT achieves the highest average performance on both backbones and is best or tied-best on every task-backbone combination. This suggests that using human data both to initialize task-relevant behavior and to retain human supervision during subsequent human-robot cotraining makes the most effective use of the transferred demonstrations.

Qualitatively, we also observe that human supervision can introduce useful motion variations absent from the robot demonstrations. On Pick Chips Can, the Robot-only policy predominantly approaches the can from the front and often knocks it over when its orientation is perturbed. With human supervision, the policy can instead adopt a side approach when appropriate, selecting different grasp motions according to the object orientation. Since this side-grasp pattern is absent from the robot demonstrations but present in the human data, the behavior provides a concrete example of how human supervision can expand the motion patterns available to the learned policy.

D. Q3: Data Efficiency

We fix 50 human demonstrations per task, vary the number of robot demonstrations over $\{0, 10, 25, 50\}$, and evaluate

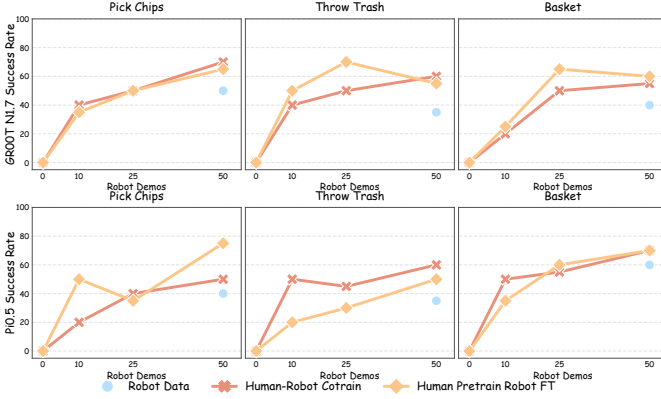


Fig. 8. **Robot-data efficiency with human supervision.** Success rate under robot demonstration budgets of $\{0, 10, 25, 50\}$ with 50 human demonstrations fixed per task. With aligned human data, success rises sharply once only a few robot demonstrations are available, and 25 already approach or match the 50-demo Robot-only baseline.

different human supervision training strategies, with Robot-only using 50 robot demonstrations as the reference (Fig. 8).

1) **Human data reduces robot data requirements:** With only 25 robot demonstrations, human-assisted training on GR00T N1.7 already matches or exceeds the 50-demo Robot-only baseline on all three tasks, effectively **reducing the required robot demonstrations by 50%**. On $\pi_{0.5}$, the same 25-demo setting remains within 5 percentage points of the 50-demo Robot-only baseline. These results show that aligned human supervision substantially improves robot-data efficiency, enabling comparable policy performance with fewer robot demonstrations.

2) **Few-shot robot data activates human priors:** With no robot demonstrations, neither human-assisted strategy achieves any task success. However, performance rises sharply once only 10 robot demonstrations are introduced. This suggests that aligned human data provides useful task and motion priors that can be activated with only a small amount of robot-specific supervision. At the same time, the failure of the human-only setting shows that human data amplifies rather than replaces robot data: robot demonstrations remain necessary to ground the policy in the target embodiment and execution domain.

The two training paradigms also exhibit different scaling behavior. Cotrain generally improves more steadily as robot data increases, whereas Human Pretrain + Robot FT is less stable. We hypothesize that finetuning exclusively on a limited robot dataset can partially overfit to the small robot distribution and weaken the diversity introduced during human pretraining, while cotraining continuously retains human supervision throughout optimization.

E. Q4: Alignment Analysis

We ablate the two alignment stages that can be removed in isolation, on Pick Chips Can and Throw Trash (Tab. III).

1) **Absolute actions fail without semantic alignment:** Replacing DexRoam’s robot-centric relative action representation with absolute retargeted poses causes the policy to fail completely: success is 0% on both tasks and under all three training strategies (Tab. III). We attribute this failure to the

TABLE III: **Ablation of alignment stages.** GR00T N1.7 success rate.

Task	Strategy	w/o Temporal Resampling	Absolute Action	DexRoam (Ours)
Pick Chips Can	Cotrain	0.65	0.00	0.70
	Pretrain + FT	0.30	0.00	0.65
	Pretrain + Cotrain FT	0.20	0.00	0.90
Throw Trash	Cotrain	0.40	0.00	0.65
	Pretrain + FT	0.50	0.00	0.55
	Pretrain + Cotrain FT	0.60	0.00	0.65

fact that absolute poses map human and robot demonstrations into a shared coordinate representation while still retaining a substantial amount of human-specific information. During training, these factors pull the supervision signal away from the robot action distribution the policy will face at deployment, so the policy learns targets entangled with human-specific configurations rather than a stable robot control semantics.

This distribution shift is further amplified over closed-loop execution. At the start of an episode, the robot’s initial observation $\mathbf{o}_0 = (\mathbf{I}_0, \mathbf{P}_0)$ lies within the distribution of the robot data, so the first predicted chunk $\mathbf{a}_{0:H-1}$ still exhibits a plausible execution trend. However, because the training distribution has been pulled toward that of human-specific absolute actions, the model biases its predicted targets toward the demonstrator’s absolute configurations and incurs a small error. Executing this error drives the next observation $\mathbf{o}_H = (\mathbf{I}_H, \mathbf{P}_H)$ off the training distribution, which in turn makes the following chunk $\mathbf{a}_{H:2H-1}$ more erroneous. Repeated over an episode, this forms a self-amplifying feedback loop that eventually drives the robot into a state from which it cannot recover, and the task fails.

Among the three strategies, Human Pretrain + Robot FT exhibits this behavior to a somewhat lesser degree under the absolute-action setting, since finetuning on robot data pulls the learned distribution partly back toward the robot’s own. The finetuning budget, however, is not sufficient to recover it fully, and the resulting policy still completes no trials.

2) **Temporal alignment improves policy learning:** Removing task-progress temporal resampling reduces the average success rate from 68.3% to 44.2%, a drop of **24.1 percentage points** (Tab. III). This degradation is caused by the different execution timescales of human and robot demonstrations: without resampling, the same task progress can correspond to different future action chunks, introducing temporally inconsistent supervision. Task-progress temporal resampling aligns human and robot trajectories at the same execution phase, removing this temporal mismatch.

F. Generalization Under Distribution Shift

We additionally evaluate two distribution shifts on Pick Chips Can — a changed background scene and an unseen instruction phrasing. As shown in Fig. 9, we observe consistent improvements under both background and instruction shifts. For background shift, Robot-only drops from 50% to 20%, while Human Pretrain + Cotrain FT decreases from 90% to 55%, maintaining the highest robustness in the novel scene. For instruction shift, strategies that retain human data during

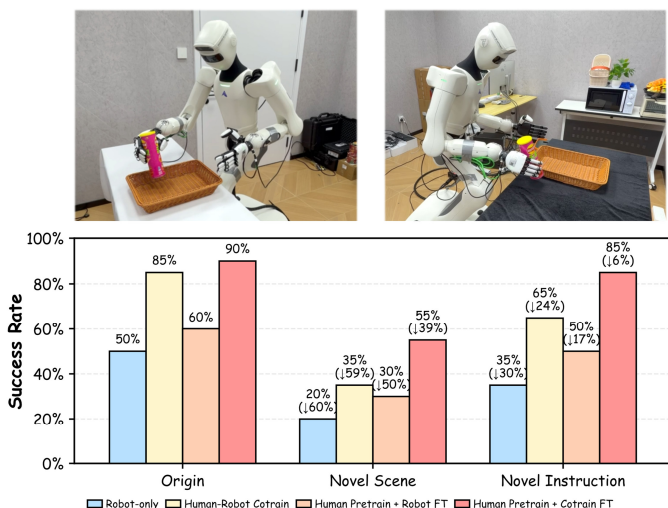


Fig. 9. **Generalization under distribution shift.** Evaluation on Pick Chips Can under the original setting, a novel scene, and an unseen instruction phrasing. The images above show the original and novel scenes. Human-assisted training degrades far less than Robot-only under both shifts, especially when human data is retained during finetuning.

finetuning, including Cotrain and Human Pretrain + Cotrain FT, remain largely stable, with only 7% and 6% relative declines, respectively. These results suggest that aligned human supervision improves robustness beyond the training distribution, particularly when human data continues to participate during policy optimization.

V. RELATED WORK

A. Mobile Manipulation

Mobile manipulation requires navigation, whole-body reachability, and object interaction to be coordinated over spatially extended tasks. Accessible platforms and data systems have progressed from Stretch and Mobile ALOHA to integrated or modular whole-body pipelines such as Tidy-Bot++, BRS, TeleMoMa, MoMa-Teleop, HoMeR, and SuperSuit [2], [3], [33]–[38]. Humanoid systems including HumanPlus, HOMIE, TWIST/TWIST2, HumanoidExo, and Teleopit further extend teleoperation and policy learning to locomotion, posture, active vision, and dexterous hands [4], [5], [39]–[42]. Beyond collection interfaces, generalist and whole-body VLA or world-action models such as GR00T N1, PanoVLA, Ψ_0 , OpenHLM, and ω_0 increasingly address language-conditioned bimanual, wheeled, and humanoid loco-manipulation [28], [31], [43]–[45]. HERMES also targets mobile bimanual dexterity but connects learned manipulation skills to a separate navigation module [46]. Despite this progress, many systems still require robot-in-the-loop collection, specialized interfaces, or substantial embodiment-specific robot data, motivating scalable robot-free demonstrations for continuous whole-body learning.

B. Human-Robot Alignment

Transferring human demonstrations to robots requires resolving differences in visual observations, morphology, action semantics, dynamics, and execution speed. Existing work

learns cross-embodiment visual or task representations [25], [26], [47], expresses behavior through transferable affordances, flow, or point trajectories [48]–[51], or translates and retargets human data into robot-compatible observations and actions [52]–[59]. For mobile whole-body learning, recent systems often make transfer tractable through discrete locomotion commands, phase-aware policies, decoupled base and manipulation actions, or end-effector and keypoint targets executed by downstream controllers [19]–[24], [60]. These abstractions improve compatibility but may omit part of the original coupled motion. DexRoam instead explicitly aligns embodiment, action semantics, and task progress while retaining a continuous, temporally uninterrupted, jointly predicted base–torso–arm–head–hand action space.

C. Human Demonstrations for Robot Learning

Human demonstrations provide diverse behavioral priors at substantially lower collection cost than robot teleoperation. Large egocentric datasets have progressed from first-person activities and hand–object interactions to paired viewpoints, metric geometry, full-body motion, and robot-oriented dexterous trajectories [7]–[9], [61]–[66]. Early work mainly extracted reusable representations or rewards, whereas portable interfaces and transfer pipelines now convert human experience into policy supervision for fixed-base, bimanual, and dexterous manipulation [10]–[18], [25], [26], [67]–[71]. Human-centric VLA and world-model pretraining further scale this idea across data sources and embodiments [72]–[76]. More recently, EMMA, EgoHumanoid, HoMMI, HuMI, BifrostUMI, HALOMI, HERMES, WARP, and OpenHLM extend human supervision to mobile or whole-body policies [19]–[24], [44], [46], [57]. Preserving mobile, bimanual, and finger-level coordination in one robot-native policy interface nevertheless remains comparatively underexplored.

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APPENDIX
THEORETICAL ANALYSIS OF STRUCTURE-PRESERVING
HUMAN-TO-ROBOT ALIGNMENT

A. Scope and Corrected Problem Formulation

This appendix provides a theoretical interpretation of the explicit alignment pipeline in Sec. III-B. The central object is a *task-conditioned trajectory transformation*, rather than a pointwise map between two pre-existing action spaces. This distinction matters because the raw human recording contains an egocentric image and a whole-body state $\mathbf{h}_t$, not a robot-semantic action. Moreover, relative actions require two adjacent poses and temporal resampling requires the complete episode.

1) *Trajectory distributions*: For a task condition c , which includes task identity and its language instruction, let

$$\tau^H = \left(\ell, \{(\mathbf{I}_i^H, \mathbf{h}_i)\}_{i=0}^{N_h-1} \right) \sim \mathbb{P}_H^c, \quad (1)$$

$$\tau^R = \left(\ell, \{(\mathbf{I}_k^R, \mathbf{p}_k^R, \mathbf{a}_k^R)\}_{k=0}^{N_r-1} \right) \sim \mathbb{P}_R^c, \quad (2)$$

Each one-step robot action is a synchronized tuple

$$\mathbf{a}_k \in \mathcal{A}_1 = \mathcal{A}^{\text{base}} \times \mathcal{A}^{\text{torso}} \times \mathcal{A}^{\text{larm}} \times \mathcal{A}^{\text{rarm}} \times \mathcal{A}^{\text{head}} \times \mathcal{A}^{\text{lhand}} \times \mathcal{A}^{\text{rhand}}. \quad (3)$$

The windowed policy target $\mathbf{A}_k = (\mathbf{a}_k, \dots, \mathbf{a}_{k+K-1})$ therefore belongs to the chunk space $\mathcal{A} = \mathcal{A}_1^K$. This makes explicit that the symbol $\mathcal{A}$ used for the policy output in Sec. II denotes a chunk space, whereas the distribution diagnostics below operate on $\mathcal{A}_1$. Thus, the human and robot distributions in (1)–(2) do not initially share an action sample space.

DexRoam constructs a task-dependent map

$$\mathcal{R}_c : \mathfrak{T}_H^c \rightarrow \mathfrak{T}_{PI}^c, \quad \mathcal{R}_c = \mathcal{R}_{\text{temp}}^c \circ \mathcal{R}_{\text{sem}} \circ \mathcal{R}_{\text{emb}}, \quad (4)$$

where $\mathfrak{T}_{PI}^c$ is the common policy-interface trajectory space. Membership means that image, proprioceptive, and action tensors have the robot policy’s format and action semantics; it does not imply that a human RGB stream lies on the support of robot-camera observations or that the entire aligned sample is a rollout of robot dynamics. The factors in (4) address embodiment, action semantics, and temporal scale, respectively. Formally, $\mathcal{R}_{\text{sem}}$ constructs a relative-action field indexed by progress and step size, and $\mathcal{R}_{\text{temp}}^c$ evaluates that field on the interpolated normalized-coordinate grid. This is an analytical interpretation of the composite, equivalent to resampling the continuous pose targets and then forming finite differences. If an implementation instead interpolates already discretized actions, its deviation from this ideal field is included in the interpolation residual of Theorem 1.

For a trajectory τ with $N(\tau) \geq K$, let $\mathcal{K}_K(\tau) = \{0, \dots, N(\tau) - K\}$ be its valid window starts and $\mathcal{W}_{K,k}(\tau) = \mathbf{z}_k = (\mathbf{I}_k, \mathbf{p}_k, \ell, \mathbf{A}_k)$. Here $\mathbf{p}_k = \mathcal{H}_P(\mathbf{p}_{0:k})$ for a fixed deterministic history constructor $\mathcal{H}_P$, including the same boundary/padding convention in both domains; $\mathcal{R}_c$ supplies the robot-format proprioceptive sequence used by this constructor for aligned human data. Because one trajectory induces several windows, the relevant object is the Markov kernel

$$\mathcal{W}_K(B | \tau) = \frac{1}{|\mathcal{K}_K(\tau)|} \sum_{k \in \mathcal{K}_K(\tau)} \mathbf{1}\{\mathcal{W}_{K,k}(\tau) \in B\}. \quad (5)$$

The full policy-example laws are consequently

$$\begin{aligned} Q_R^c(B) &= \int \mathcal{W}_K(B | \tau) d\mathbb{P}_R^c(\tau), \\ \tilde{Q}_H^c(B) &= \int \mathcal{W}_K(B | \mathcal{R}_c(\tau)) d\mathbb{P}_H^c(\tau). \end{aligned} \quad (6)$$

This convention weights episodes equally and then samples a valid start uniformly; another training sampler simply induces a different, explicitly specified kernel.

For the analysis, fix a common reference probability measure ν_c on latent task progress $s \in [0, 1]$. Let $\mathbf{V}_{\nu_c}(\cdot | \tau)$ draw $s \sim \nu_c$ and return the phase-tagged pair $(s, \mathbf{a}_\tau(s))$, where $\mathbf{a}_\tau(s)$ is the one-step action obtained from the interpolated policy-interface trajectory. Define

$$\begin{aligned} \overline{Q}_R^c(B) &= \int \mathbf{V}_{\nu_c}(B | \tau) d\mathbb{P}_R^c(\tau), \\ \overline{Q}_H^c(B) &= \int \mathbf{V}_{\nu_c}(B | \mathcal{R}_c(\tau)) d\mathbb{P}_H^c(\tau). \end{aligned} \quad (7)$$

These laws live on $[0, 1] \times \mathcal{A}_1$ and hence admit regular conditional action laws given s . Their untagged marginals $\overline{Q}_R^{c,a}$ and $\overline{Q}_H^{c,a}$ are population analogues of the *fully aligned* one-step whole-body vectors assessed in Figs. 5 and 6; an absolute, pre-semantic law would be defined by applying the same sampling kernel before the last two alignment stages. Both are distinct from the chunk variable $\mathbf{A}$ used by the policy.

2) *Structure-preserving admissible class*: The transformation-specific restrictions define an admissible class $\mathfrak{R}_{SP}^c$:

- **Continuous-valued motion**: every output coordinate remains in the continuous one-step space $\mathcal{A}_1$ and no finite motion vocabulary or quantizer is introduced.
- **Order-preserving time**: one common, increasing progress coordinate is used for the entire episode.
- **Synchronized whole-body coupling**: all components in (3) are sampled at the same progress value, placed in one joint action vector, and packed into the same chunk.

Prediction by one unified policy, with no phase-specific switching at deployment, is the associated policy-architecture requirement; it is not a property of the preprocessing map $\mathcal{R}_c$ itself. These requirements preserve robot-relevant motion structure, rather than all raw human degrees of freedom: planar root projection, two-axis head control, and morphology-dependent hand retargeting intentionally quotient out motion that the target embodiment cannot execute.

A useful analytical design criterion is therefore

$$\begin{aligned} \min_{\mathcal{R}_c \in \mathfrak{R}_{SP}^c} \quad & \mathbb{E}_{s \sim \nu_c} \left[D_{\mathcal{A}_1} \left(\overline{Q}_H^c(\mathbf{a} | s, c), \overline{Q}_R^c(\mathbf{a} | s, c) \right) \right], \quad (8) \\ \text{subject to} \quad & \text{range}(\mathcal{R}_c) \subseteq \mathfrak{T}_{PI}^c, \quad \varepsilon_{\text{kin}}(\mathcal{R}_c) \leq \bar{\varepsilon}_{\text{kin}}. \end{aligned}$$

Here ε_{kin} measures retargeting/IK residual on the geometric channels and is infinite if no feasible target-embodiment solution exists; $\bar{\varepsilon}_{\text{kin}}$ is a prescribed feasibility tolerance, and $D_{\mathcal{A}_1}$ is any chosen discrepancy between probability laws on the normalized one-step action space. The conditional laws are those induced by (7), and ν_c fixes how task progress is weighted. Equation (8) is a constrained *design desideratum*, not an additional objective optimized by DexRoam. The

method constructs one explicit member of $\mathcal{R}_{\text{SP}}^c$; MMD, SWD, and PCA are post-hoc diagnostics. Conditioning on progress also makes the intended semantics explicit. Matching only the unordered action marginal is not sufficient, as shown next.

B. Why Distribution-Only Alignment Is Insufficient

Proposition 1 (Marginal matching is non-identifying). Let $\mathbf{a} = (\mathbf{a}_0, \dots, \mathbf{a}_{N-1})$ be any whole-body action trajectory and

$$\mu_{\mathbf{a}} = \frac{1}{N} \sum_{t=0}^{N-1} \delta_{\mathbf{a}_t} \quad (9)$$

its unordered empirical action distribution. For every permutation σ of $\{0, \dots, N-1\}$, $\mu_{\mathbf{a} \circ \sigma} = \mu_{\mathbf{a}}$. Consequently, any discrepancy computed solely on one-step action marginals assigns zero distance to the original and permuted sequences, although their temporal transitions and task-stage order can be different.

Proof: Reindexing a finite sum does not change the measure:

$$\frac{1}{N} \sum_t \delta_{\mathbf{a}_{\sigma(t)}} = \frac{1}{N} \sum_j \delta_{\mathbf{a}_j}. \quad (10)$$

Thus every distribution discrepancy that depends only on this measure is unchanged. Choosing a nonmonotone permutation can alter adjacency relations $(\mathbf{a}_t, \mathbf{a}_{t+1})$, so equality of action marginals cannot identify temporal structure. If each body component is permuted independently, its component marginal is also unchanged while synchronous cross-component pairing is destroyed. A discrepancy on the *joint* whole-body vector, as used in the paper, can detect same-step coupling but still cannot detect episode order. ■

The same limitation can coexist with arbitrarily accurate Wasserstein matching and explicit quantization. On a compact action set, let q_ϵ map every action to its nearest point in a finite ϵ -net. Coupling $\mathbf{a}$ with $q_\epsilon(\mathbf{a})$ gives

$$W_p(P, (q_\epsilon)_\# P) \leq \epsilon \quad (11)$$

for every $p \geq 1$, even though q_ϵ is finite-valued and noninjective. Hence a small distribution distance alone does not certify continuous, ordered, or fine-grained motion. This is why (8) restricts the alignment map before assessing distribution compatibility.

C. Structural Properties of the DexRoam Map

We now state the properties supplied by the three explicit alignment stages. Let the continuous robot-compatible pose curve produced by embodiment retargeting be

$$\mathbf{q}(s) = (\mathbf{X}^b(s), \mathbf{X}^c(s), \mathbf{X}^l(s), \mathbf{X}^r(s), \mathbf{u}^h(s), \mathbf{u}^{lh}(s), \mathbf{u}^{rh}(s)), \quad (12)$$

where $s \in [0, 1]$ denotes task progress, $\mathbf{X}^b \in \text{SE}(2)$ is the root/chassis pose, $\mathbf{X}^c, \mathbf{X}^l, \mathbf{X}^r \in \text{SE}(3)$ are the chest and two wrist targets, and the $\mathbf{u}$ terms collect the direct head and hand joint targets. Mechanical feasibility and retargeting fidelity are properties of $\mathcal{R}_{\text{emb}}$; the results below do not assume that the non-executable part of the human motion can be preserved.

1) *Root-frame gauge invariance:* **Proposition 2 (Invariance to the world planar gauge).** Let $\mathbf{G} \in \text{SE}(3)$ be a fixed rigid transformation that preserves gravity and the ground plane; equivalently, its rotation obeys $\mathbf{R}_G \mathbf{z}_H = \mathbf{z}_H$ and its translation lies in the ground plane. Suppose

$$\|\Pi_G^W \mathbf{f}_{\text{pelvis},t}\|_2 \geq \eta > 0 \quad (13)$$

so that the projected pelvis-forward direction is well-defined. If all human world poses are changed by ${}^W\mathbf{T}'_{J,t} = \mathbf{G} {}^W\mathbf{T}_{J,t}$, then

$${}^W\mathbf{T}'_{H,t} = \mathbf{G} {}^W\mathbf{T}_{H,t}, \quad (14)$$

$${}^{H'}\mathbf{T}'_{J,t} = {}^H\mathbf{T}_{J,t}, \quad (15)$$

$$\Delta \mathbf{T}_t^{H'} = \Delta \mathbf{T}_t^H. \quad (16)$$

Thus the root-relative upper-body poses and the local chassis increments are independent of global ground-plane origin and yaw.

Proof: Because $\mathbf{G}$ preserves gravity, $\Pi_G \mathbf{R}_G = \mathbf{R}_G \Pi_G$. Hence both the projected pelvis origin and projected forward axis transform by $\mathbf{G}$, while normalization is valid by (13). This proves (14). Direct substitution gives

$$\begin{aligned} {}^{H'}\mathbf{T}'_{J,t} &= (\mathbf{G} {}^W\mathbf{T}_{H,t})^{-1} \mathbf{G} {}^W\mathbf{T}_{J,t} \\ &= ({}^W\mathbf{T}_{H,t})^{-1} {}^W\mathbf{T}_{J,t} = {}^H\mathbf{T}_{J,t}, \end{aligned} \quad (17)$$

and the same cancellation in two consecutive root frames proves (16). ■

Absolute world poses do not have this invariance. Proposition 2 formalizes why the pelvis-centric construction removes a scene-coordinate nuisance before human and robot supervision are mixed.

2) *Relative action semantics:* **Proposition 3 (Completeness modulo a constant gauge).** Let $\mathbb{G}$ be any matrix Lie group and let $\mathbf{X}_{0:N} = (\mathbf{X}_0, \dots, \mathbf{X}_N) \in \mathbb{G}^{N+1}$. The anchored relative encoding

$$\Psi(\mathbf{X}_{0:N}) = (\mathbf{X}_0, \{\Delta \mathbf{X}_t\}_{t=0}^{N-1}), \quad \Delta \mathbf{X}_t = \mathbf{X}_t^{-1} \mathbf{X}_{t+1}, \quad (18)$$

is bijective. If the anchor $\mathbf{X}_0$ is omitted, the increments determine the trajectory exactly up to a common left multiplication.

Proof: Given the encoding, recursively set

$$\mathbf{X}_{t+1} = \mathbf{X}_t \Delta \mathbf{X}_t. \quad (19)$$

This reconstructs one and only one anchored trajectory, proving bijectivity. For any constant $\mathbf{C} \in \mathbb{G}$,

$$(\mathbf{C}\mathbf{X}_t)^{-1}(\mathbf{C}\mathbf{X}_{t+1}) = \mathbf{X}_t^{-1}\mathbf{X}_{t+1}, \quad (20)$$

so unanchored increments are unchanged along a common left orbit. Conversely, two sequences with the same increments differ only through their initial left factor, by recursion in (19). ■

Proposition 3 applies componentwise to the product group $\text{SE}(2) \times \text{SE}(3)^3$ in (12). If each current product pose $\mathbf{X}_t$ is deterministically recoverable from proprioception by the known forward kinematics and root convention, and the controller uses the same right-side composition $\mathbf{X}_{t+1} = \mathbf{X}_t \Delta \mathbf{X}_t$, each relative action uniquely specifies the next geometric target. The action-semantic stage therefore does not quantize

or replace the fine-grained sequence with a motion primitive; it removes a constant coordinate/initial-configuration gauge. More specifically, under the nuisance model

$$\mathbf{X}_t^d = \mathbf{C}_d \mathbf{Z}_t, \quad d \in \{H, R\}, \quad (21)$$

the human and robot relative motions both equal $\mathbf{Z}_t^{-1} \mathbf{Z}_{t+1}$ even when $\mathbf{C}_H \neq \mathbf{C}_R$. This exact statement covers constant left nuisances. Nonlinear morphology effects, IK error, and discarded head roll remain part of embodiment-retargeting error and are not removed by (20) alone.

3) *Continuity*: **Proposition 4 (Continuous-valued aligned motion)**. Assume that (13) holds, the retargeted targets in (12) are continuous, the implemented rotation parameterization and SLERP branch are continuous on each interpolation segment, and base yaw is unwrapped. Then the geometric aligned actions are samples of a piecewise-continuous curve in $\mathcal{A}_1$. In particular, the listed geometric operators introduce no finite action quantizer.

Proof: Under the stated regularity condition, ground projection and vector normalization are continuous. Group inversion and multiplication are smooth. The retargeted target curve is continuous by assumption, linear position interpolation is continuous, and the selected SLERP and rotation-coordinate branches are continuous by assumption. Their composition with the relative encoding is therefore continuous on each interpolation interval. Concatenating all body components does not change continuity. ■

This statement concerns geometric actions. Images and episode-state signals are matched by nearest neighbor in the implementation and are not claimed to be continuous. Likewise, the absence of deployment phase switching follows from the unified-policy interface, not from Proposition 4. If the underlying position curve is twice differentiable and the largest human progress gap is Δ_h , linear interpolation has the standard error bound

$$\sup_s \|\hat{\mathbf{p}}(s) - \mathbf{p}(s)\|_2 \leq \frac{\Delta_h^2}{8} \sup_s \|\mathbf{p}''(s)\|_2, \quad (22)$$

with an analogous $O(\Delta_h^2)$ local bound inside a geodesically convex neighborhood when the rotation curve is twice co-variantly differentiable with bounded covariant acceleration. For merely continuous motion, the interpolation error instead vanishes according to its modulus of continuity as $\Delta_h \rightarrow 0$.

4) *Temporal order and synchronized coupling*: **Proposition 5 (Common progress preserves synchronization)**. Let

$$\gamma(s) = (\gamma_1(s), \dots, \gamma_m(s)), \quad s \in [0, 1], \quad (23)$$

be the joint whole-body curve, and let $\phi : [0, 1] \rightarrow [0, 1]$ be a strictly increasing continuous bijection. Then:

- 1) γ and $\gamma \circ \phi$ have identical endpoints and the same oriented path;
- 2) the order of any two events at $s_1 < s_2$ is unchanged; and
- 3) for any components i, j , their synchronous coupling graph

$$\Gamma_{ij}(\gamma) = \{(\gamma_i(s), \gamma_j(s)) : s \in [0, 1]\} \quad (24)$$

is unchanged by the common reparameterization.

Proof: Strict monotonicity preserves order and endpoints. Surjectivity gives

$$\begin{aligned} \Gamma_{ij}(\gamma \circ \phi) &= \{(\gamma_i(\phi(t)), \gamma_j(\phi(t))) : t \in [0, 1]\} \\ &= \Gamma_{ij}(\gamma). \end{aligned} \quad (25)$$

The same argument applied to the full tuple proves equality of the oriented joint path. ■

Proposition 5 concerns the ideal continuous reparameterization. DexRoam evaluates every component on the same finite normalized-coordinate grid $r_k = (k - 1)/(\bar{N}_r - 1)$. The retained samples preserve index order, endpoints, and synchronization at every retained coordinate, and they do not introduce component-specific time warps. A finite sample set does not, however, equal the full path or preserve an event that falls strictly between samples. If γ is L_γ -Lipschitz in a product metric and the largest grid gap is Δ_r , every point on the continuous path is within $L_\gamma \Delta_r / 2$ of a retained waypoint. This guarantee does not recover information discarded by a noninjective retargeting map.

D. Alignment Error and Distribution Discrepancy

We next quantify the role of temporal resampling. Let $\mathbf{X}_d(s) \in \text{SE}(2) \times \text{SE}(3)^3$ denote the geometric part of (12), and let $\mathbf{u}_d(s)$ collect the direct joint targets. For $d \in \{H, R\}$ and $0 \leq s < s + h \leq 1$, define

$$\mathbf{b}_d(s, h) = \chi(\mathbf{X}_d(s)^{-1} \mathbf{X}_d(s + h), \mathbf{u}_d(s + h)) \quad (26)$$

as the vectorized hybrid whole-body robot action at progress s with progress increment h . Here χ parameterizes the geometric relative transforms and concatenates the direct head and hand targets at the next waypoint; the inverse and product are taken componentwise on the product group.

1) *Regularity assumptions*: The demonstrations need not be observed in pairs. For analysis only, assume there exists a coupling π over same-task latent instances, human and robot episodes, and valid normalized-coordinate starts (r_H, r_R) . Let $\psi_d : [0, 1] \rightarrow [0, 1]$ be a continuous, strictly increasing phase map with $\psi_d(0) = 0$ and $\psi_d(1) = 1$, mapping normalized episode coordinate to latent task progress, and write

$$\Delta\psi_d(r; h) = \psi_d(r + h) - \psi_d(r). \quad (27)$$

The following assumptions isolate the discrepancies addressed by DexRoam.

- **Progress-proxy residuals**. Under π , compared starts and increments obey

$$|\psi_H(r_H) - \psi_R(r_R)| \leq \delta_0, \quad (28)$$

$$|\Delta\psi_H(r_H; h) - \Delta\psi_R(r_R; h_R) - (h - h_R)| \leq \delta_h, \quad (29)$$

for $h \in \{h_H, \bar{h}_R\}$ at starts for which all displayed increments are valid. Thus δ_0 and δ_h are direct cross-domain residuals, not separate per-domain errors.

- **Embodiment residual**. After embodiment and semantic alignment, actions at equal progress and equal progress increment satisfy

$$\|\mathbf{b}_H(s, h) - \mathbf{b}_R(s, h)\|_2 \leq \varepsilon_E. \quad (30)$$

- **Local regularity.** On the compact feasible motion set,

$$\|\mathbf{b}_d(s, h) - \mathbf{b}_d(s', h')\|_2 \leq L_s|s - s'| + L_h|h - h'|. \quad (31)$$

The first assumption is substantive. Uniform normalized time is a valid task-progress proxy when episodes traverse the same ordered behavior at different overall speeds. If their internal stage speeds differ arbitrarily, δ_0 or δ_h can be large and length matching alone cannot guarantee semantic phase alignment. For example, if each domain’s phase estimate has pointwise error at most δ and the same normalized start is compared, one generally obtains only $\delta_0 \leq 2\delta$ and $\delta_h \leq 4\delta$; the definitions above avoid silently dropping these cross-domain factors.

Theorem 1 (Temporal-alignment error bound). Let

$$h_H = \frac{1}{N_h - 1}, \quad h_R = \frac{1}{N_r - 1}, \quad \bar{h}_R = \frac{1}{\bar{N}_r - 1}. \quad (32)$$

Here $\bar{N}_r$ is the task-specific reference length chosen by DexRoam (derived from the empirical mean robot length), so $\bar{h}_R$ is a fixed reference step and need not equal $\mathbb{E}[h_R]$. Under π , set

$$\begin{aligned} \mathbf{a}_R &= \mathbf{b}_R(\psi_R(r_R), \Delta\psi_R(r_R; h_R)), \\ \mathbf{a}_H^{\text{pre}} &= \mathbf{b}_H(\psi_H(r_H), \Delta\psi_H(r_H; h_H)), \\ \mathbf{a}_H^{\text{ideal}} &= \mathbf{b}_H(\psi_H(r_H), \Delta\psi_H(r_H; \bar{h}_R)), \end{aligned} \quad (33)$$

and assume the implemented interpolant obeys $\|\mathbf{a}_H^{\text{align}} - \mathbf{a}_H^{\text{ideal}}\|_2 \leq \varepsilon_{\text{int}}$. The superscript “pre” denotes a robot-semantic relative action before the temporal stage; it is not the absolute “Raw distribution” plotted in Figs. 5–6. Define

$$\rho_j = \left(\mathbb{E}_\pi \|\mathbf{a}_H^j - \mathbf{a}_R\|_2^2 \right)^{1/2}, \quad j \in \{\text{pre}, \text{align}\}. \quad (34)$$

Then

$$\rho_{\text{pre}} \leq \varepsilon_E + L_s\delta_0 + L_h\delta_h + L_h(\mathbb{E}_\pi |h_H - h_R|^2)^{1/2}, \quad (35)$$

$$\begin{aligned} \rho_{\text{align}} &\leq \varepsilon_E + L_s\delta_0 + L_h\delta_h + \varepsilon_{\text{int}} \\ &\quad + L_h(\mathbb{E}_\pi |\bar{h}_R - h_R|^2)^{1/2}. \end{aligned} \quad (36)$$

Proof: Write $s_d = \psi_d(r_d)$ and $\hat{h}_d = \Delta\psi_d(r_d; h_d)$. Insert the robot-domain action at $(s_H, \hat{h}_H)$. The triangle inequality, (30), and (31) give

$$\begin{aligned} \|\mathbf{a}_H^{\text{pre}} - \mathbf{a}_R\|_2 &\leq \varepsilon_E + L_s|s_H - s_R| + L_h|\hat{h}_H - \hat{h}_R| \\ &\leq \varepsilon_E + L_s\delta_0 + L_h\delta_h + L_h|h_H - h_R|. \end{aligned} \quad (37)$$

Minkowski’s inequality under π proves (35). After resampling, use $h = \bar{h}_R$ in (29) and add the interpolation residual; the same argument proves (36). ■

The temporal stage therefore replaces the variable human normalized-coordinate step h_H by the fixed reference step $\bar{h}_R$; the remaining rate term is the variation of robot episode steps around that reference. The bound becomes smaller when that variation and interpolation error are smaller than the original cross-domain rate mismatch. This is a conditional statement about the controlled error terms, not a claim that every finite-sample discrepancy must decrease. For a K -step chunk with the ordinary stacked Euclidean norm, the right-hand sides increase by at most a factor $\sqrt{K}$; with the mean per-step chunk metric, they are unchanged.

2) *Consequences for Population SWD and MMD:* **Corollary 1 (Metric bounds under progress coupling).** Suppress c and let $\bar{Q}_R^a$ and $\bar{Q}_H^a$ denote the one-step action marginals in (7). Assume the coupling used in Theorem 1 induces a coupling $\pi_a \in \Pi(\bar{Q}_R^a, \bar{Q}_H^a)$ whose root-mean-square action cost is at most ρ_{align} . Let $S : \mathcal{A}_1 \rightarrow \mathbb{R}^m$ be the common action-normalization map and assume S is L_{norm} -Lipschitz. Write $\bar{Q}_{d,S}^a = S_\# \bar{Q}_d^a$. Then the population order-two sliced Wasserstein distance satisfies

$$\text{SWD}_2(\bar{Q}_{R,S}^a, \bar{Q}_{H,S}^a) \leq W_2(\bar{Q}_{R,S}^a, \bar{Q}_{H,S}^a) \leq L_{\text{norm}}\rho_{\text{align}}. \quad (38)$$

For the Gaussian kernel

$$k_\sigma(x, y) = \exp\left(-\frac{\|x - y\|_2^2}{2\sigma^2}\right), \quad (39)$$

the unsquared maximum mean discrepancy also obeys

$$\text{MMD}_{k_\sigma}(\bar{Q}_{R,S}^a, \bar{Q}_{H,S}^a) \leq \frac{L_{\text{norm}}\rho_{\text{align}}}{\sigma}. \quad (40)$$

Proof: Pushing π_a through (S, S) gives a valid coupling whose root-mean-square cost is at most $L_{\text{norm}}\rho_{\text{align}}$. The Wasserstein distance is the infimum over all such couplings. Every one-dimensional projection is 1-Lipschitz, which yields $\text{SWD}_2 \leq W_2$. If φ_σ is the Gaussian-kernel feature map, then

$$\begin{aligned} \|\varphi_\sigma(x) - \varphi_\sigma(y)\|_{\mathcal{H}}^2 &= 2\left(1 - e^{-\|x - y\|_2^2/(2\sigma^2)}\right) \\ &\leq \frac{\|x - y\|_2^2}{\sigma^2}. \end{aligned} \quad (41)$$

Taking expectations under a coupling and then the infimum gives $\text{MMD}_{k_\sigma} \leq W_1/\sigma \leq W_2/\sigma$, which proves the stated normalized-coordinate bound. ■

Corollary 1 isolates the contribution of temporal alignment after fixing the hybrid action semantics; Propositions 2–3 explain why the embodiment and semantic stages remove particular nuisance coordinates. Together they provide a mechanism consistent with, but not a proof of, the total empirical reduction in Fig. 6. The displayed bounds concern population SWD_2 with unit-direction averaging and unsquared Gaussian MMD under a common normalization. Finite-sample estimators add sampling error; MMD squared, another kernel or bandwidth rule, or another SWD aggregation changes the reported form or constants. Consequently, the exact normalization, estimator, projection, and PCA protocol must be stated from the experiment implementation rather than inferred from this theory.

E. Why Aligned Human Data Can Improve Robot Learning

The previous results concern the action marginal. Policy learning instead depends on the full robot-native example $z = (\mathbf{I}, \mathbf{P}, \ell, \mathbf{A})$. We now give a target-risk bound on this full space. For readability, suppress c and write Q_H for the aligned human policy-example law $\bar{Q}_H^c$ and Q_R for the robot law. Define the cotraining mixture

$$Q_\alpha = (1 - \alpha)Q_R + \alpha Q_H, \quad 0 \leq \alpha \leq 1, \quad (42)$$

and, for supervised loss $g_\theta(z)$,

$$\mathfrak{R}_D(\theta) = \mathbb{E}_{z \sim D}[g_\theta(z)]. \quad (43)$$

The loss-class discrepancy is

$$d_{\mathcal{L}}(Q_R, Q_H) = \sup_{\theta \in \Theta} |\mathfrak{R}_{Q_R}(\theta) - \mathfrak{R}_{Q_H}(\theta)|. \quad (44)$$

Theorem 2 (Robot-domain risk of mixture training). Let $\hat{\theta}$ be an ε_{opt} -approximate empirical-risk minimizer on samples from Q_{α} . Suppose that, on an event of probability at least $1 - \delta$,

$$\sup_{\theta \in \Theta} |\mathfrak{R}_{Q_{\alpha}}(\theta) - \hat{\mathfrak{R}}_{Q_{\alpha}}(\theta)| \leq \varepsilon_N. \quad (45)$$

If $\theta_R^* \in \arg \min_{\theta \in \Theta} \mathfrak{R}_{Q_R}(\theta)$, then, on the same event,

$$\begin{aligned} \mathfrak{R}_{Q_R}(\hat{\theta}) &\leq \mathfrak{R}_{Q_R}(\theta_R^*) + 2\varepsilon_N \\ &\quad + 2\alpha d_{\mathcal{L}}(Q_R, Q_H) + \varepsilon_{\text{opt}}. \end{aligned} \quad (46)$$

Proof: By linearity of (42),

$$|\mathfrak{R}_{Q_R}(\theta) - \mathfrak{R}_{Q_{\alpha}}(\theta)| \leq \alpha d_{\mathcal{L}}(Q_R, Q_H). \quad (47)$$

Applying (47), the uniform bound, and approximate empirical optimality in sequence gives

$$\begin{aligned} \mathfrak{R}_{Q_R}(\hat{\theta}) &\leq \mathfrak{R}_{Q_{\alpha}}(\hat{\theta}) + \alpha d_{\mathcal{L}} \\ &\leq \hat{\mathfrak{R}}_{Q_{\alpha}}(\hat{\theta}) + \varepsilon_N + \alpha d_{\mathcal{L}} \\ &\leq \hat{\mathfrak{R}}_{Q_{\alpha}}(\theta_R^*) + \varepsilon_N + \alpha d_{\mathcal{L}} + \varepsilon_{\text{opt}} \\ &\leq \mathfrak{R}_{Q_{\alpha}}(\theta_R^*) + 2\varepsilon_N + \alpha d_{\mathcal{L}} + \varepsilon_{\text{opt}} \\ &\leq \mathfrak{R}_{Q_R}(\theta_R^*) + 2\varepsilon_N + 2\alpha d_{\mathcal{L}} + \varepsilon_{\text{opt}}, \end{aligned} \quad (48)$$

where the arguments of $d_{\mathcal{L}}$ are omitted only for compactness. ■

The discrepancy in (46) can be controlled by familiar distribution metrics under standard regularity assumptions. If every g_{θ} belongs to an RKHS $\mathcal{H}_k$ with $\|g_{\theta}\|_{\mathcal{H}_k} \leq B$, then

$$d_{\mathcal{L}}(Q_R, Q_H) \leq B \text{MMD}_k(Q_R, Q_H). \quad (49)$$

If every loss is instead L_g -Lipschitz under the metric on policy examples, the Kantorovich–Rubinstein duality gives

$$d_{\mathcal{L}}(Q_R, Q_H) \leq L_g W_1(Q_R, Q_H). \quad (50)$$

These are statements about the full joint policy-example distributions, not only their action marginals. The action MMD/SWD in the experiments diagnoses one marginal aspect of the gap; it does not upper-bound the joint discrepancy. Visual appearance, proprioception, and execution dynamics can leave a residual domain discrepancy.

For comparison, a robot-only estimator has the analogous excess-risk bound $2\varepsilon_{N_R} + \varepsilon_{\text{opt}}^R$. Ignoring differences in optimization error, the mixture guarantee is tighter whenever

$$\varepsilon_N + \alpha d_{\mathcal{L}}(Q_R, Q_H) < \varepsilon_{N_R}. \quad (51)$$

Equation (51) exposes the relevant trade-off: additional human trajectories can reduce the estimation term ε_N , while alignment is designed to reduce part of the domain-bias term. The guarantee improves only to the extent that it reduces the full $d_{\mathcal{L}}(Q_R, Q_H)$. It also explains why aligned human data need not replace robot data: with $\alpha = 1$, residual observation and dynamics gaps can still make the bound loose. Theorem 2 directly describes cotraining and the final cotraining phase of Human Pretrain + Cotrain FT. The effect of human-only pretraining followed by robot-only finetuning additionally

depends on representation transfer and optimization and is not claimed by this bound.

The uniform-convergence term in Theorem 2 is an explicit assumption, not a claim that overlapping windows are independent. Here N must be interpreted as an episode-level or otherwise justified effective sample size; a concrete bound for correlated windows requires an appropriate dependent-data generalization result.

F. Closed-Loop Error Accumulation

The action-semantic ablation in Sec. IV-E shows that mismatched target semantics can be destructive during rollout. A standard conditional stability calculation makes one possible accumulation mechanism explicit.

Corollary 2 (Propagation of action error). Suppose the closed-loop transition map satisfies

$$d_x(F(x, a), F(x', a')) \leq L_x d_x(x, x') + L_a \|a - a'\|_2. \quad (52)$$

Let $e_t = d_x(x_t, x_t^*)$ and let $\eta_t = \|a_t - a_t^*\|_2$. If both sequences follow $x_{t+1} = F(x_t, a_t)$ and $x_{t+1}^* = F(x_t^*, a_t^*)$, and $x_0 = x_0^*$, then

$$e_T \leq L_a \sum_{t=0}^{T-1} L_x^{T-1-t} \eta_t. \quad (53)$$

Proof: Equation (52) gives $e_{t+1} \leq L_x e_t + L_a \eta_t$. Repeated substitution with $e_0 = 0$ yields (53). ■

Thus persistent absolute-pose or phase-inconsistent supervision can accumulate over a long horizon, especially when $L_x \geq 1$. Root-relative semantics and progress alignment are designed to reduce identifiable sources of η_t , but (53) is a conditional stability bound; it does not by itself prove that either representation produces a particular action error or guarantee task success. Equation (53) is written at the low-level action step; the same proof applies to chunk execution after replacing F by the induced K -step transition and a_t by $\mathbf{A}_t$.

G. Interpretation and Limits of the Guarantees

The exact results are the non-identifiability of marginal matching, planar gauge invariance, completeness of anchored relative actions, and preservation of order and synchronous pairing under a common monotone progress map. Under the stated shared-progress and regularity assumptions, temporal resampling also replaces the human step-size mismatch by a fixed robot-reference-rate term in an action-error bound, which upper-bounds population SWD/MMD under the stated coupling; the mixture-risk result then quantifies when extra aligned samples can improve a robot-domain guarantee.

These guarantees are deliberately conditional. The normalized episode index is only a proxy for semantic progress, with cross-domain residuals δ_0 and δ_h , and only robot-relevant structure modulo embodiment constraints is preserved. The analysis proves neither global optimality of $\mathcal{R}_c$, strict reduction of every empirical distance, nor higher task success. Executability, distribution-gap reduction, and policy improvement remain the empirical validation of the favorable regime characterized above.